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Multi-lobe balloon for cryoablation

Abstract

A cryotreatment catheter for treating tissue. The catheter may include an outer elongate body, a balloon treatment element coupled to the distal portion of the elongate body with a plurality of balloon lobes radially arranged around the outer elongate body, an inner elongate body rotatably movable within the lumen of the outer elongate body, and a fluid delivery lumen located within the lumen of the outer elongate body and at least partially within the lumen of the inner elongate body. The fluid delivery lumen may be branched at a distal portion into a plurality of linear segments, each linear segment being in fluid communication with one of the plurality of balloon lobes. Each of the balloon lobes may be inflated independently of each other by the linear segments of the fluid delivery lumen.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a divisional application of Patent Application Ser. No. 17/325,307, filed May 20, 2021, now U.S. Pat. No. 11,986,231, which is a continuation of Patent Application Ser. No. 16/368,277, filed Mar. 28, 2019, now U.S. Pat. No. 11,045,245, issued Jun. 29, 2021, which is a continuation of Patent Application Ser. No. 15/988,431, filed May 24, 2018, now U.S. Pat. No. 10,299,848, issued May 28, 2019, which is a continuation of patent application Ser. No. 14/944,870, filed Nov. 18, 2015, now U.S. Pat. No. 10,058,371, issued Aug. 28, 2018, the entire contents of each of which are incorporated herein by reference.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) n/a

TECHNICAL FIELD

(2) The present invention relates to an adjustable treatment device for accommodating a variety of pulmonary vein morphologies and a method of ablating tissue using the same.

BACKGROUND

(3) A cardiac arrhythmia is a condition in which the heart's normal rhythm is disrupted. Certain types of cardiac arrhythmias, such as paroxysmal atrial fibrillation, may originate from an arrhythmogenic focus in or close to the pulmonary veins. Recent studies of pulmonary vein (PV) morphology showed that there is a wide variability in, for example, PV diameters at the ostia between the position of the veins within the same patient's heart (right superior PV, left superior PV, right inferior PV, and left inferior PV). These studies also showed that the diameter and cross-sectional area of the left superior PV are gender dependent, being significantly larger in men than in women.

(4) Many types of cardiac arrhythmia may be treated by various ablation methods, including cryoablation. Data analysis of cryotherapy procedures showed that the use of cryoballoon catheters in combination with focal or radiofrequency (RF) catheters is common. Generally speaking, the focal or RF catheters are used to access sections of the PVs at the ostia that are inaccessible using only the cryoballoon catheter or that still exhibit conduction of aberrant electrical signals after cryoballoon ablation. The average rate of using such touch-up methods for three consecutive years

was about 11% (2011, 2012, and 2013).

(5) Given the variation in PV morphology between patients of different genders and, indeed, within a single patient, means that treating arrhythmia using a one-size-fits-all cryoballoon catheter without the need for one or more touch-up procedures is nearly impossible. As the need for focal or RF catheter ablation following cryoablation poses a safety risk to the patient and increases procedure time, it is desirable to provide a cryoablation device that can be adjusted to accommodate a variety of PV morphologies, such as PV diameter, cross-sectional area, shape, etc.

SUMMARY

(6) The present invention advantageously provides a device and system that is adjustable for accommodating a variety of pulmonary vein morphologies and a method of ablating tissue using the same. A cryotreatment catheter may generally include an elongate body including a distal portion, a proximal portion, and lumen therebetween, and a balloon treatment element coupled to the distal portion of the elongate body, the balloon treatment element including a plurality of balloon lobes radially arranged around the elongate body. For example, the balloon treatment element may include at least six balloon lobes. The distal portion of the elongate body may include a first plurality of apertures and a second plurality of apertures located distal to the first plurality of apertures, each of the first and second pluralities of apertures being radially arranged around the elongate body and corresponding to one of the plurality of balloon lobes. Each of the plurality of balloon lobes may include a first aperture that is radially aligned with one of the plurality of first apertures of the elongate body, and a second aperture that is radially aligned with one of the plurality of second apertures of the elongate body, the second aperture being located distal to the first aperture. The elongate body may be an outer elongate body, and the cryotreatment catheter may further include an inner elongate body including a distal portion, a proximal portion, a lumen therebetween, and a plurality of apertures radially arranged around the inner elongate body. Each of the plurality of apertures of the inner elongate body may correspond to one of the first plurality of apertures of the outer elongate body. Further, the distal portion of the inner elongate body may define a distal end, the distal end of the inner elongate body being distal to the first plurality of apertures of the outer elongate body. The cryotreatment catheter further may include a delivery lumen at least partially located within the inner elongate body. The delivery lumen may include a proximal portion located within the inner elongate body, and a branched distal portion located within the outer elongate body distal to the distal end of the inner elongate body, the branched distal portion including a plurality of linear segments, each of the plurality of linear segments being in fluid communication with a corresponding one of the plurality of balloon lobes. Each of the linear segments may include a distal tip portion that has a delivery aperture, each of the distal tip portions extending from a corresponding linear segment at an approximately 90° angle, each of the distal tip portions extending through a corresponding one of the second plurality of apertures of the outer elongate body and a corresponding second aperture of one of the plurality of balloon lobes. The lumen of the outer elongate body may be configured to be in fluid communication with a vacuum source and a fluid reservoir.

(7) A cryotreatment catheter may include: an outer elongate body including a distal portion, a proximal portion, and a lumen extending between the distal portion and the proximal portion, the distal portion defining a distal end; a balloon treatment element coupled to the distal portion of the elongate body, the balloon treatment element including a plurality of balloon lobes radially arranged around the outer elongate body; an inner elongate body located within and rotatably and/or linearly movable within the lumen of the outer elongate body, the inner elongate body including a distal portion, a proximal portion, and a lumen extending between the distal portion and the proximal portion, the distal portion defining a distal end that is located proximal to the distal end of the outer elongate body; a fluid delivery lumen located within the lumen of the outer elongate body and at least partially within the lumen of the inner elongate body. The outer elongate body may include a first plurality of apertures and a second plurality of apertures located distal to

the first plurality of apertures, each of the first and second pluralities of apertures being radially arranged around the elongate body and corresponding to one of the plurality of balloon lobes. each of the plurality of balloon lobes may include a first aperture that is radially aligned with one of the plurality of first apertures of the elongate body, and a second aperture that is radially aligned with one of the plurality of second apertures of the elongate body, the second aperture being located distal to the first aperture. The inner elongate body may include a plurality of apertures radially arranged around the inner elongate body, each of the apertures of the inner elongate body corresponding to one of the first plurality of apertures of the outer elongate body. The plurality of balloon lobes may include at least six balloon lobes. The fluid delivery lumen may include: a proximal portion located within the inner elongate body; a branched distal portion located within the outer elongate body, the branched distal portion including a plurality of linear segments, each of the plurality of linear segments being in fluid communication with a corresponding one of the plurality of balloon lobes; and a divergence point between the proximal portion and the branched distal portion, the divergence point being distal to the distal end of the inner elongate body. Each of the linear segments may include a distal tip portion that has a delivery aperture, each of the distal tip portions extending from a corresponding linear segment at an approximately 90° angle, each of the distal tip portions extending through a corresponding one of the second plurality of apertures of the outer elongate body and a corresponding second aperture of one of the plurality of balloon lobes. Each of the outer elongate body and the inner elongate body may include a longitudinal axis, the longitudinal axis of the inner elongate body being coaxial with the longitudinal axis of the outer elongate body, and the inner elongate body is configured to obstruct the first plurality of apertures of the outer elongate body when the inner elongate body is rotated and/or linearly moved along its longitudinal axis.

(8) A cryotreatment catheter may include: an outer elongate body including a longitudinal axis, a distal portion, a proximal portion, and a lumen extending between the distal portion and the proximal portion, the distal portion defining a distal end, the outer elongate body further including a first plurality of apertures and a second plurality of apertures located distal to the first plurality of apertures; a balloon treatment element coupled to the distal portion of the elongate body, the balloon treatment element including a plurality of balloon lobes radially arranged around the outer elongate body, each of the plurality of balloon lobes having an attachment spine and a tissue contact surface, a first plurality of apertures, and a second plurality of apertures located distal to the first plurality of apertures, the first plurality of apertures of the balloon lobes being radially aligned with the first plurality of apertures of the outer elongate body and the second plurality of apertures of the balloon lobes being radially aligned with the second plurality of apertures of the outer elongate body; an inner elongate body located within and rotatably movable within the lumen of the outer elongate body, the inner elongate body including a longitudinal axis that is coaxial with the longitudinal axis of the outer elongate body, a distal portion, a proximal portion, a lumen extending between the distal portion and the proximal portion, and a plurality of apertures at the distal portion, the distal portion defining a distal end that is located proximal to the distal end of the outer elongate body, and the plurality of apertures of the inner elongate body being configured to be radially aligned with the first plurality of apertures of the outer elongate body and the first plurality of apertures of the balloon lobes, the inner elongate body being configured to obstruct the first plurality of apertures of the outer elongate body when the inner elongate body is rotated along its longitudinal axis; and a fluid delivery lumen including a proximal portion, a branched distal portion, and a divergence point therebetween, the divergence point being located distal to the plurality of apertures of the inner elongate body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete understanding of the present invention, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:
- (2) FIG. 1 shows an exemplary system including a treatment catheter having an adjustable balloon treatment element;
- (3) FIG. 2 shows a balloon lobe;
- (4) FIG. 3A shows a cross-sectional view of a fluid delivery lumen within the treatment catheter of FIG. 1;
- (5) FIG. 3B shows a close-up view of a portion of the fluid delivery lumen;
- (6) FIG. 4 shows a distal cross-sectional view of a first embodiment of the balloon treatment element with all lobes being fully inflated;
- (7) FIG. 5 shows a distal cross-sectional view of a second embodiment of the balloon treatment element with all lobes being fully inflated;
- (8) FIG. 6A shows a general inflation position of an inner lumen relative to the balloon treatment element;
- (9) FIG. 6B shows a precision inflation position of the inner lumen relative to the balloon treatment element;
- (10) FIG. 6C shows an ablation position of the inner lumen relative to the balloon treatment element;
- (11) FIG. 7 shows a distal cross-sectional view of a first example of an asymmetrical inflation of the balloon treatment element;
- (12) FIG. 8 shows a distal cross-sectional view of a second example of an asymmetrical inflation of the balloon treatment element;
- (13) FIGS. 9A-9D show configurations of an inner elongate body within a treatment catheter during a cryotreatment procedure; and
- (14) FIGS. 10 and 11 show cross-sectional views of thermoplastic tubes within a mold for forming the multi-lobed balloon treatment element.

DETAILED DESCRIPTION

(15) The present invention advantageously provides a device and system that is adjustable for accommodating a variety of pulmonary vein morphologies and a method of ablating tissue using the same. Referring now to the drawing figures in which like reference designations refer to like elements, an exemplary system including a cryoablation catheter having an adjustable cryoballoon treatment element is shown in FIG. 1 and generally designated as “**10**.” The device components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Moreover, while certain embodiments or figures described herein may illustrate features not expressly indicated on other figures or embodiments, it is understood that the features and components of the system and devices disclosed herein are not necessarily exclusive of each other and may be included in a variety of different combinations or configurations without departing from the scope and spirit of the invention.

(16) The system **10** may generally include a treatment catheter **12** and a control unit **14** in communication with the treatment device **12**. The treatment device **12** may be, for example, a cryoballoon catheter and the control unit **14** may be configured for use with a cryotreatment procedure. The cryotreatment catheter **12** may include a balloon treatment element **18** that includes a plurality of balloon lobes **20**. As is described in more detail below, each of the balloon lobes **20** may be inflated and deflated independently of each other.

(17) Referring to the treatment catheter **12** in more detail, the catheter **12** may include an elongate body **24** passable through a patient's vasculature and/or positionable proximate to a tissue region for diagnosis or treatment, such as a catheter, sheath, or intravascular introducer. The elongate body **24** may include a proximal portion **26**, a distal portion **28**, and a longitudinal axis **30**, and may further include one or more lumens disposed within the elongate body **24** that provides mechanical, electrical, and/or fluid communication between the elongate body proximal portion **26** and the elongate body distal portion **28**. For example, the elongate body **24** may include a main lumen **32**. The balloon treatment element **18** may be coupled to the distal portion **28** of the elongate body **24**, with the plurality of balloon lobes **20** being radially arranged about the elongate body **24**.

(18) Continuing to refer to FIG. **1**, the control unit **14** may include a fluid supply including one or more reservoirs **36** for one or more coolants, cryogenic refrigerants, or the like, an exhaust or scavenging system for recovering or venting expended fluid for reuse or disposal (including, for example, a recovery reservoir and vacuum pump), as well as various control mechanisms. The control unit **14** may also include an additional fluid supply including a reservoir **38** containing a non-coolant liquid, gas, or combination liquid and gas used to inflate the balloon lobes **20**, which fluid may be referred to as "inflation fluid." In addition to providing an exhaust function for the inflation fluid and/or refrigerant reservoirs, the control unit **14** may also include pumps, valves, controllers or the like to recover and/or re-circulate fluid delivered to various fluid pathways of the catheter. A vacuum pump **40** in the control unit **14** may create a low-pressure environment in one or more conduits within the catheter so that fluid is drawn into the conduits/lumens of the elongate body **24**, away from the distal portion **28** and toward the proximal portion **26** of the elongate body **24**.

(19) The control unit **14** may also include one or more components for the manual, automatic, and/or semi-automatic regulation of the system, such as a computer **42** having one or more processors **44** for executing one or more algorithms for the automatic or semi-automatic regulation of the catheter **12** before, during, and after an ablation procedure. For example, the one or more processors **44** may be programmable to at least partially inflate and at least partially deflate the plurality of balloon lobes **20**, to regulate temperature of the balloon treatment element **18**, and/or to receive and interpret mapping or sensor signals from the cryotreatment catheter **12** or another device used as part of a medical procedure. The control unit **14** may also include, for example, a proportional valve that regulates inflation of the balloon treatment element **18** during the transition between the inflation phase and the ablation phase. Although various system components may be shown and described herein as being within the control unit **14**, the term "control unit" as used herein refers to any system component other than the cryotreatment catheter and other devices that are passed into the patient to perform the medical procedure, regardless of whether the components are physically located within the control unit **14**.

(20) Referring now to FIG. **2**, the balloon treatment element **18** may be, for example, a cryotreatment element. As shown in the figures, the balloon treatment element **18** may include a plurality of balloon lobes **20** radially arranged about the elongate body **24**. As a non-limiting example, the balloon treatment element **18** may include six balloon lobes, although fewer or more balloon lobes **20** may be included. Each of the balloon lobes **20** may include a proximal portion **48**, a distal portion **50**, and an attachment spine **52** proximate the elongate body **24**, a tissue contact surface **54** opposite the attachment spine **52**, and each lobe **20** may define an inner chamber **56**. Further, each balloon lobe **20** may include two lateral surfaces **58** that are in contact with the lateral surfaces **58** of adjacent lobes **20**. Each balloon lobe **20** may be shaped approximately like a section of an orange and the attachment spine **52** of each lobe **20** may be affixed to the distal portion **28** of the elongate body **24**. As a non-limiting example, each attachment spine **52** may be affixed proximally to the elongate body distal portion **28** and distally to a distal tip **59** using an adhesive, chemical or thermal bonding, or other suitable attachment means (for example, as shown in FIG. **1**). Alternatively, each attachment spine **52** may be affixed proximally and distally to the elongate

body **24** (for example, as shown in FIGS. **3A** and **9A-9D**). Further, each attachment spine **52** may include a first (proximal) aperture **60** and a second (distal) aperture **62**. Likewise, the distal portion **28** of the elongate body **24** may include a plurality of first (proximal) apertures **64**, each of which corresponding to an adjacent first aperture **60** of a balloon lobe attachment spine **52**. For example, if the balloon treatment element **18** includes six balloon lobes **20** each having a first aperture **60**, the distal portion **28** of the elongate body **24** may include six first apertures **64** that are radially arranged about the elongate body **24** longitudinal axis and configured to match up with or be in fluid communication with each of the balloon lobe first apertures **60**. The distal portion **28** of the elongate body **24** may further include a plurality of second (distal) apertures **66**, each of which corresponding to an adjacent second aperture **62** of a balloon lobe attachment spine **52**. For example, if the balloon treatment element **18** includes six balloon lobes **20** each having a second aperture **62**, the distal portion **28** of the elongate body **24** may include six second apertures **66** that are radially arranged about the elongate body **24** longitudinal axis **30** and configured to match up with or be in fluid communication with each of the balloon lobe second apertures **62**. Each of the apertures disclosed herein may be a round or substantially round hole in the wall of the elongate body **24** or the balloon lobe attachment spine **52**. However, it will be understood that the apertures may be of any size or configuration that allows the passage of fluid therethrough. Further, the diameter of each aperture may be sized to provide a desired fluid flow rate.

(21) Referring now to FIGS. **3A** and **3B**, the elongate body **24** may define a main lumen **32**, which may function as both an inflation lumen and an exhaust lumen. The catheter **12** may also include an inner elongate body **70** defining an inner lumen **72**, and the inner elongate body **70** may be rotatable and/or longitudinally (linearly) movable within the main lumen **32** of the elongate body **24**, which may be referred to as being an outer elongate body **24** relative to the inner elongate body **70**. The inner elongate body **70** may also include a plurality of apertures **74** that may be radially and longitudinally aligned with the first apertures **64** of the elongate body **24** and the first apertures **60** of the balloon lobes **20**, such that rotation and/or longitudinal movement of the inner elongate body **70** in at least one direction may selectively obscure/obstruct or unobscure/unobstruct the first apertures **64** of the elongate body **24** and the first aperture **60** of each of the balloon lobes **20**. Additionally or alternatively, the inner elongate body **70** may be advanced or retracted within the main lumen **32** to selectively obscure/obstruct or unobscure/unobstruct the first apertures **64**, **60** of the elongate body **24** and the balloon lobes **20**. The elongate body **24** and the inner elongate body **70** may each define a proximal portion and a distal portion, with each distal portion defining a distal end. The distal end **76** of the inner elongate body **70** may be at a location that is proximal to the distal end **78** of the elongate body **24** (as shown in FIG. **3A**).

(22) The catheter **12** may also include a delivery lumen **82** that includes a proximal portion **84** and a distal portion **86**. At least a portion of the proximal portion **84** of the delivery lumen **82** may be located within the inner elongate body lumen. The distal portion **86** of the delivery lumen **82** may be branched into a plurality of linear segments **88**, with each linear segment **88** being in fluid communication with one of the plurality of balloon lobes **20** through the second apertures **66** of the elongate body **24** and the second aperture **62** of the balloon lobe **20**. Further, the proximal portion **84** of the delivery lumen **82** may end, and the distal portion **86** of the delivery lumen **82** may begin, at a branch or divergence point **90** location that is distal to the distal end **76** of the inner elongate body **70** and distal to the elongate body first apertures **64** (as shown in FIG. **3A**).

(23) As is shown in detail in FIG. **3B**, each linear segment **88** of the distal portion **86** of the delivery lumen **82** may include a distal tip portion that includes a fluid delivery segment **96**. The fluid delivery segment **96** extend from the linear segment **88** at an approximately 90° angle ($\pm 5^\circ$), forming an L shape. Thus, each branch of the delivery lumen distal portion **86** may be referred to herein as being “L shaped.” Each of the fluid delivery segments **96** may extend at least partially through a corresponding second aperture **66** of the elongate body **24** and a second aperture **62** of a balloon lobe **20** to be in fluid communication with the lobe inner chamber **56**. The tip of the fluid

delivery segment **96** may define or include a delivery aperture **98**, through which an inflation and/or cryogenic fluid may be delivered to the inner chamber of the balloon lobe **20**. The second apertures **66** of the elongate body **24** and the second apertures **62** of the balloon lobes **20** may be sized and configured such that they are able to receive the fluid delivery segment **96** of each linear segment but without allowing inflation fluid and/or cryogenic refrigerant to leak back into the main lumen **32**. For example, the fluid delivery segment **96** may be friction fit and/or coupled within the second aperture **66** of the elongate body **24** and the second aperture **62** of the balloon lobes **20** (such as with an adhesive, chemical or thermal bonding, or other suitable means). A proximal portion of each linear segment **88** may pass through a sheath or lumen to form a bundle, or may be coupled to one another to form a bundle, and this bundle may be referred to as the proximal portion **84** of the delivery lumen **82**.

(24) Referring now to FIGS. **4** and **5**, distal cross-sectional views of two embodiments of the balloon treatment element are shown. In the first embodiment shown in FIG. **4**, the balloon treatment element **18** may include only a plurality of balloon lobes **20**. In the second embodiment shown in FIG. **5**, however, the balloon treatment element **18** may also include an outer balloon **102** that surrounds the plurality of balloon lobes **20**. The outer balloon **102** may be composed of a material that is more compliant than the material from which the plurality of balloon lobes **20** are composed, and the outer balloon **102** may help smooth the transitional curves between the balloon lobes **20** and create a smooth tissue contact surface that covers all or at least substantially all of the balloon treatment element **18**. The outer balloon **102** may include a distal neck and a proximal neck (not shown) that are affixed or coupled to the distal tip **59** and the distal portion **28** of the elongate body **24**, respectively. Alternatively, the outer balloon **102** may be coupled to the elongate body **24** in another suitable manner.

(25) Although not shown, the treatment catheter **12** may also be configured for use with other energy modalities, such as radiofrequency energy, microwave energy, laser energy, ultrasound energy, electroporation energy, and the like. Therefore, it will be understood that the balloon treatment element **18** may also include one or more electrodes and/or other energy delivery elements. As a non-limiting example, the balloon treatment element **18** may include one or more fiber sensors located within the balloon lobes **20** and/or between the balloon lobes **20** and the outer balloon for temperature assessment of the ablated tissue and/or strain assessment. Further, the one or more fiber sensors may be used for hyperspectral assessment of the targeted tissue. The medical system may also include one or more sensors to monitor the operating parameters throughout the system, including for example, pressure, temperature, flow rates, volume, or the like in the control unit **14**, and/or the catheter **12**. For example, the catheter **12** may further include one or more temperature and/or pressure sensors (not shown) proximate and/or within the balloon treatment element **18** for monitoring, recording or otherwise conveying measurements of conditions within the medical device or the ambient environment at the distal portion of the medical device. The sensor(s) may be in communication with the control unit **14** for initiating or triggering one or more alerts or therapeutic delivery modifications during operation of the medical device.

(26) The treatment catheter **12** may include a handle **110** coupled to the proximal portion **26** of the elongate body **24** and/or the inner elongate body **70**. The handle **110** may include one or more actuation elements **112**, such as sliders, levers, or knobs, for manipulating the elongate body **24**, inner elongate body **70**, and/or additional components of the medical device. For example, the handle **110** may include a knob that is in mechanical communication with the inner lumen, such that rotation of the knob may produce a similar rotation of the inner lumen within the main lumen **32** of the elongate body **24**. Additionally or alternatively, rotation of the inner elongate body **70** may be controlled automatically or semi-automatically by the control unit **14**. For example, the handle **110** may include a geared motor controlled by the control unit **14** to transmit movement to the elongate body **70**. Alternatively or additionally, the handle **110** may include a slider, lever, or other actuation element that is also in mechanical communication with the inner elongate body **70**.

such that activation of the actuation element selectively advances or retracts (that is, longitudinally moves) the inner elongate body **70** in at least one longitudinal direction and/or rotates the inner elongate body **70** in a clockwise or counterclockwise direction. The handle **110** may further include circuitry for identification and/or use in controlling of the catheter or another component of the system **10**. Additionally, the handle **110** may be provided with a fitting for receiving a guide wire that may be passed into the main lumen **32** or a guidewire lumen (not shown).

(27) Referring now to FIGS. **6A-6C**, various positions of the inner elongate body relative to the balloon treatment element are shown. For simplicity, the delivery lumen **82** and linear segments **88** are not shown as it is in FIGS. **4** and **5**. As described above, the inner elongate body **70** may include a plurality of apertures **74**, each of which corresponding to an elongate body first aperture **64** and a balloon lobe first aperture **60**. For example, if the balloon treatment element **18** includes six lobes, the inner elongate body **70** may include six apertures. As shown in FIG. **6A**, the balloon treatment element **18** may be inflated when the inner elongate body **70** is rotated such that the apertures **74** of the inner elongate body **70** are aligned with or correspond to the first apertures **64** of the elongate body **24** and the first apertures **60** of the balloon lobes **20**. In this configuration, an inflation fluid may be delivered to the inner chambers **56** of the balloon lobes **20** through the main lumen **32** of the elongate body **24**, through the apertures **74** of the inner elongate body **70**, through the first apertures **64** of the elongate body **24**, and through the first apertures **60** of the balloon lobes **20**. As shown in FIG. **6B** and described in more detail below, each balloon lobe **20** may be further selectively inflated when the inner elongate body **70** is rotated such that the inner elongate body **70** apertures **74** are not aligned with or correspond to the first apertures **60** of the balloon lobes **20** and the first apertures **64** of the elongate body **24**. In this configuration, inflation fluid within the balloon lobes **20** is prevented from evacuating through the first apertures **60**, **64** and further inflation fluid may be precisely delivered to one or more lobes **20** by one or more linear segments **88** of the delivery lumen **82**. Finally, as shown in FIG. **6C**, the inflation fluid may be evacuated and replaced by a cryogenic refrigerant as it flows without affecting the inflation level of each balloon lobe **20** when the inner elongate body **70** is rotated such that the inner elongate body apertures **74** are only partially aligned with or corresponding to the first apertures **60** of the balloon lobes **20** and the first apertures **64** of the elongate body **24**. In this configuration, the flow rate and/or injection pressure of the cryogenic refrigerant being delivered by the delivery lumen **82** to each balloon lobe **20** may be balanced by the evacuation rate of the fluid from the balloon lobes **20** through the inner elongate body apertures **74**.

(28) Referring now to FIGS. **7** and **8**, examples of asymmetrical inflation of the balloon treatment element are shown. As is described in more detail below, each balloon lobe **20** may be inflated independently of the others in order to ensure contact between the balloon treatment element **18** and a variety of target tissue configurations. As is shown in FIG. **7**, for example, the balloon treatment element **18** may be initially inflated and positioned at a pulmonary vein ostium, or at least partially inserted into the pulmonary vein. Then, each balloon lobe **20** may be precisely inflated as needed, depending on the morphology of the pulmonary vein. This may be particularly useful when the pulmonary vein has a non-circular or irregular cross section on which it is difficult to produce a circumferential lesion with traditional, non-lobed cryoballoons. Likewise, the balloon treatment element **18** may be inflated to contact a lateral area of target tissue. As shown in FIG. **8**, for example, the balloon treatment element **18** may be initially inflated and positioned proximate an area of target tissue on a cardiac wall. Then, the balloon lobes that are not in contact with the target tissue may not be inflated at all and the balloon lobes that are in contact with the target tissue may be inflated using the delivery lumen **82** and the linear segments corresponding to each of the balloon lobes to be inflated. Deflating or not inflating the balloon lobes that are not in contact with the target tissue may reduce the warming effect of surrounding blood on the balloon treatment element **18**.

(29) Referring now to FIGS. **9A-9D**, configurations of the inner elongate body within the treatment

catheter during a cryotreatment procedure are shown. The treatment catheter **12** may first be navigated through the patient's vasculature to a target treatment site. As a non-limiting example, the treatment catheter **12** may be navigated to the left atrium of the patient's heart, to a location at which the balloon treatment element **18** is in contact with or proximate a pulmonary vein ostium. Once the treatment catheter **12** is at the desired location, the balloon lobes **20** may be inflated during a primary inflation cycle. As shown in FIG. 9A, during the primary inflation cycle, the inner elongate body **70** may be positioned such that the inner elongate body apertures **74** are open (that is, such that the inner elongate body apertures **74** are fully aligned with the first apertures **64** of the elongate body **24** and the first apertures **60** of each balloon lobe **20**). An inflation fluid may then be delivered from the inflation fluid reservoir **38** through the main lumen **32** of the elongate body **24**, from where it passes through the apertures **74** of the inner elongate body **70**, the first apertures **64** of the elongate body **24**, and the first apertures **60** of the balloon lobes **20**. The inflation fluid may continue to be delivered this way until all of the balloon lobes **20** are inflated and the balloon treatment element **18** has an at least approximately circular cross-sectional shape. Any suitable inflation fluid may be used, such as room-temperature nitrous oxide (N.sub.2O) vapor. As is shown in FIG. 9B, the inner elongate body **70** may then be rotated either clockwise or counterclockwise such that the inner elongate body apertures **74** are closed (that is, no longer fully aligned with the first apertures **64** of the elongate body **24** and the first apertures **60** of each balloon lobe **20**). In this position, fluid flows from the main lumen **32** into the balloon lobes **20**, and evacuation of fluid from the balloon lobes **20** into the main lumen **32**, may be prevented. Although not shown, the inner elongate body **70** may alternatively be moved longitudinally (advanced or retracted) within the elongate body **24** so the inner elongate body apertures **74** are no longer open or aligned with the first apertures **64** of the elongate body **24** and the balloon lobes **20**.

(30) As is shown in FIG. 9C, inflation fluid may be delivered from the inflation fluid reservoir **38** through the delivery lumen **82** and into each of the linear segments **88** at the distal portion of the delivery lumen **82** in a secondary inflation cycle. The inflation fluid may then pass through the delivery apertures **98** of the fluid delivery segments **96** and into the inner chambers **56** of the balloon lobes **20**. Due to the closed configuration of the inner elongate body apertures **74**, fluid passing into the balloon lobes **20** may continue to inflate, and therefore increase the size of, each of the balloon lobes to which the inflation fluid is delivered. As a non-limiting example, the pulmonary vein may have an asymmetric morphology. In order to ensure contact between the entire or at least substantially the entire circumference of the balloon treatment element **18**, one or more balloon lobes may be further inflated using the secondary inflation cycle so that extend farther from the catheter **12** longitudinal axis than other balloon lobes. In other words, the outer surface of the balloon at the widest portion of some balloon lobes may be located at a greater radial distance from the catheter longitudinal axis than others. Thus, these larger balloon lobes may bridge the gap between the balloon treatment element **18** and the wall of the pulmonary vein. Inflation fluid may be delivered to all or fewer than all of the balloon lobes **20** during the secondary inflation cycle.

(31) Once the balloon treatment element **18** has the desired configuration and while the injection has started, the inner elongate body **70** may be rotated clockwise or counterclockwise until the inner elongate body apertures **74** are partially open (that is, such that the inner elongate body apertures **74** are partially aligned with the first apertures **64** of the elongate body **24** and the first apertures **60** of the balloon lobes **20**), which defines the transition phase. Then, the elongate body apertures **74** are fully open once the ablation has reached its normal flow and/or pressure. The cryogenic refrigerant may be delivered from a cryogenic refrigerant reservoir **36** through the delivery lumen **82** and into the balloon lobes **20**, as the inflation fluid was delivered in the secondary inflation cycle. During this transitional phase between inflation and ablation, cryogenic refrigerant may be injected into the balloon lobes **20** and may gradually replace the inflation fluid, which may be evacuated through the first balloon lobe apertures **60** and into the main lumen **32**. Additionally, the main lumen **32** may be in fluid communication with a vacuum source **40**, which

may facilitate evacuation of inflation fluid and cryogenic refrigerant from the balloon lobes **20**. The differential injection pressures between the balloon lobes **20** will maintain the multi-lobe balloon shape during ablation.

(32) If the treatment catheter **12** is navigated to other treatment locations, such as a wall of the left atrium, the balloon lobes that are in contact with the target tissue may be inflated using only the delivery lumen **82** as described above for the secondary inflation cycle. Further, the inner elongate body **70** may be positioned such that the inner elongate body apertures **74** are closed. In this manner, fewer than all of the balloon lobes may be inflated.

(33) Referring now to FIGS. **10** and **11**, a method of manufacturing a multi-lobed balloon treatment element is shown. Specifically, a lateral cross-sectional view of thermoplastic tubes **114** within a mold **116** for forming the multi-lobed balloon treatment element **18** is shown in FIG. **10**, and a distal cross-sectional view of the same is shown in FIG. **11**. A plurality of extruded thermoplastic tubes **114**, one for each balloon lobe, may be initially bonded to each other in a radial array to form a parison **118** around a core **120** (as shown in FIG. **11**) using an adhesive, thermal bonding, laser welding, or the like. The parison **118** may be positioned within a mold **116** that is shaped similar to the inflated balloon treatment element **18**. Each thermoplastic tube **114** may then be expanded using compressed air/gas to conform to the shape of the mold **116**.

(34) It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described herein above. In addition, unless mention was made above to the contrary, it should be noted that all of the accompanying drawings are not to scale. A variety of modifications and variations are possible in light of the above teachings without departing from the scope and spirit of the invention, which is limited only by the following claims.

Claims

1. A method of manufacturing a treatment element, the method comprising the steps of: positioning a plurality of thermoplastic tubes around a core; bonding the plurality of thermoplastic tubes in an array to form a parison around the core; positioning the bonded plurality of thermoplastic tubes in a mold; and expanding the bonded plurality of thermoplastic tubes to conform to the shape of the mold to form the treatment element to include a plurality of lobes, where each lobe of the plurality of lobes corresponds to one of the plurality of thermoplastic tubes.
 2. The method of claim 1, further comprising the step of extruding the plurality of thermoplastic tubes.
 3. The method of claim 1, wherein the bonding of the plurality of thermoplastic tubes is in a radial array to form the parison around the core.
 4. The method of claim 1, wherein the plurality of thermoplastic tubes are bonded using an adhesive.
 5. The method of claim 1, wherein the plurality of thermoplastic tubes are bonded using thermal bonding or laser welding.
 6. The method of claim 1, wherein compressed air is used to expand the bonded plurality of thermoplastic tubes.
 7. The method of claim 1, wherein the treatment element is a balloon.
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